

ECON 1740 — The American Economy

One-Page Class Sheets, Classes 1–25

Companion to the overview memo and syllabus. One page per class. Each sheet follows the same template — a thesis, a hook, a killer chart, the economics, the American-institutions connection, a discussion question, a modern echo, and readings — so that any instructor can pick up a class and run with it. Full 75-minute teaching notes for every class exist separately (cLass-outLines.md), along with a compiled chart deck.

The course thesis (Day 1, returned to throughout): America’s economic system is a **growth machine built from institutions**. It protects rights, rewards investment, connects markets, attracts people, generates ideas, scales firms, and corrects failures. Its history is both a story of extraordinary prosperity and a story of conflict over who participates, who gains, who loses, and who governs the system.

The four acts:

Part	Title (Classes)	The question
I	The Growth Miracle (1–6)	What is growth, why does it matter, and why did it emerge first in the Anglo-American world?
II	Building the American Growth Machine (7–10)	How did the U.S. build a continental market, an invention system, and organizations capable of scale?
III	Mobilizing People and Expanding Opportunity (11–18)	Who got access to the growth machine?
IV	Governing Capitalism’s Consequences (19–25)	How does America govern capitalism’s side effects? Five tools — rules (20), provision (21), insurance and transfers (22), stabilization (23), pricing harms (24) — bookended by the two open questions: globalization (19) and technology (25).

Class 1 — The American Economy in 25 Charts

Course overview — the trailer · *Institution family: all four*

For most of human history, ordinary people were poor, sick, and short-lived; this course explains how Americans became the richest people in history — and asks whether it can last.

The hook. Six charts go up, one at a time — GDP since 1500, life expectancy, productivity, immigration, the debt, AI investment. Then the question that runs the whole course: **which of these do you not believe?** Every hand that goes up is doing economics: interrogating evidence. By December, students will know how to attack — and defend — every panel on the screen.

The killer chart. The six-panel dashboard — GDP per capita since 1500, life expectancy, labor productivity, the foreign-born share, federal debt, and AI investment — one preview panel per theme. Then the promise: twenty-five classes, twenty-five charts, each one a fact that demands an explanation.

The economics. - Economics starts with facts — charts before theories. - Growth compounds — small rate differences become huge income differences. - Institutions are rules — and rules can be studied like machines. - Every chart hides a comparison — learning to ask “compared to what?”

The American institution. Day 1 states the thesis: America’s economy is a growth machine built from institutions — property rights, courts, patents, schools, immigration, finance, antitrust, the Fed. The rest of the course takes the machine apart one component at a time. Day 1 also plants the recruiting flag: a “what economists actually do” slide — Goldin on women’s careers, Chetty mapping the American Dream, Autor on trade shocks — this is the toolkit, and this course is the free trial.

Discussion. Index cards: which chart surprised you most? Which one do you *not* believe? (The skeptical cards seed the measurement-and-causality thread that runs through every class.)

Modern echo. Every panel on the dashboard is a live 2020s policy fight — growth, housing, immigration, debt, AI. Students leave knowing the course is about their world, not the past.

Read. The syllabus; Our World in Data, “Economic Growth” (browse the charts before Class 2).

Class 2 — GDP, Living Standards, and Welfare: Why Growth Matters

Part I — The Growth Miracle · *Institution family: growth-enabling*

Economic growth is one of the most important facts in human history — and while GDP is not the same as welfare, real income growth is tightly connected to almost everything people care about.

The hook. Would you rather be a king in 1600 or an ordinary American today? The king had servants; you have antibiotics, anesthesia, and a supermarket. Make it concrete with a walk through an American home in 1870 — no plumbing, no electric light, laundry as a full workday, half the household budget spent on food — versus the same home in 1970. For almost all of human history, ordinary people were poor, sick, and short-lived; then, in a handful of places, everything changed — and the United States became the leading example.

The killer chart. *The hockey stick* — GDP per capita since 1500 for the US, UK, China, India, and the world — shows millennia of flatline, then a vertical takeoff, with the U.S. line out front. *Life expectancy and world poverty* charts show growth buying survival itself: extreme poverty collapsing as incomes rise.

The economics. - Real vs. nominal GDP — correcting money for prices. - Compounding — 1% vs. 3% growth means 1.6× vs. 4.4× income in 50 years. - $GDP = C + I + G + NX$ — what gets counted, what doesn't. - Mean vs. median — averages can hide people.

The American institution. This class establishes the American economy itself as the course's central case study: of all the places that escaped stagnation, the U.S. went about as far and as fast as anyone — richest large economy, frontier of innovation, magnet for migrants. Every American institution studied later answers the question posed here: why did growth happen, and why here?

Discussion. France looks much poorer than the U.S. in GDP but works fewer hours and lives longer — which country is actually better off?

Modern echo. Today's debates over inequality and "GDP versus well-being" turn on exactly this class's distinction between production and welfare.

Read. Stevenson & Wolfers (2008), "Economic Growth and Subjective Well-Being" (anchor); Deaton, *The Great Escape* (accessible).

Class 3 — The Recipe for Growth: Capital, Labor, Ideas, and Institutions

Part I — The Growth Miracle · *Institution family: growth-enabling*

Growth is like a recipe: some comes from more ingredients — workers, machines, schooling — but the biggest long-run gains come from using those ingredients better.

The hook. GDP is produced, not given. "Productivity" sounds boring, but it is the main reason your wages and living standards will be higher than your grandparents' — the

difference between a shovel and an excavator, a paper ledger and AI. Why can one hour of work today produce so much more than in 1947?

The killer chart. *The price of light, 1300–2023* (Fouquet–Pearson data; pedagogy follows Nordhaus) — the price of lighting fell roughly 16,000-fold from candles to LEDs: productivity made visceral in a single line. Supported by *U.S. labor productivity, 1947–present* — output per hour rising relentlessly, the single fact behind rising wages — and the *BLS decomposition* splitting growth into “more inputs” versus “better recipe.”

The economics. - Production function — ingredients combine to produce output. - Capital deepening — more tools per worker. - Ideas (TFP) — nonrival recipes anyone can reuse. - Complementarity — electricity did little until factories were redesigned.

The American institution. America grew by having every ingredient at once: land, immigrant labor, mass schooling and land-grant universities, deep capital markets, and an innovation ecosystem — Bell Labs, universities, Silicon Valley. These are the institutions the rest of the course examines one by one.

Discussion. If America’s population doubled tomorrow, would America be richer?

Modern echo. AI needs chips, data centers, electricity, and firms — complementarity explains why new technology is slow to show up in productivity, and why ideas may be getting harder to find.

Read. Romer (1990), “Endogenous Technological Change” (anchor); Bloom, Jones, Van Reenen & Webb (2020), “Are Ideas Getting Harder to Find?” (accessible).

Class 4 — Liberty and the Rules of the Game

Part I — The Growth Miracle · *Institution family: growth-enabling*

Modern growth required both a new moral idea — individuals who own themselves, their property, and their conscience — and the legal machinery that made those rights credible enough to invest in.

The hook. Play a simple game, live: you have capital — invest it, or hide it? If the king can seize whatever you build, you hide it, and the economy stays poor forever. This class asks what had to change — in beliefs and in law — before ordinary people would build, lend, and trade with strangers. It is not a Western Civ lecture; it is an economics lecture about why credibility creates wealth.

The killer chart. *English government borrowing costs before and after 1688* — the North–Weingast evidence that constraining the Crown made lenders believe they would be repaid, collapsing interest rates and mobilizing the capital that financed the Industrial Revolution. Framed by *From subject to citizen* — a 600-year rights timeline from Magna Carta (1215)

through Locke, the First Amendment, and the 13th Amendment: rights are historically new, cumulative, and hard-won.

The economics. - Self-ownership — owning your labor underlies wages and migration. - Credible commitment — limits on arbitrary power unlock investment. - Property rights as a bundle — use, income, sale, liability. - Prices as information — markets coordinate self-interested strangers.

The American institution. The American colonies inherited the English package — common law, contract, religious dissent — and would soon codify it more fully than anywhere on earth (next class). Religious toleration belongs in the story as an institutional innovation: Europe spent 150 years killing over belief; toleration turned pluralism into an economic asset. Honest caveat: rights were initially partial — the story of the course is their expansion.

Discussion. The invest-or-hide game, played live — then: are markets mainly a technology for greed, or a technology for cooperation among strangers?

Modern echo. Economic-freedom and property-rights indexes still correlate strongly with income across countries today.

Read. Magna Carta (Clause 39) and Locke (short excerpts); North & Weingast (1989), “Constitutions and Commitment” (anchor); North, “Institutions” (*JEP* 1991, accessible).

Class 5 — The Constitution as an Economic Document

Part I — The Growth Miracle · *Institution family: growth-enabling*

The Constitution is, among other things, an economic charter: it created the largest free-trade zone on earth, national protection for ideas and contracts, credible public credit, and a marketplace for religion — the operating system the American growth machine still runs on.

The hook. Read the Constitution the way an economist reads a startup’s charter. In 1787, thirteen quarreling mini-countries had trade wars, worthless debt, and competing currencies; they merged into a single market with one commerce policy, one money, uniform bankruptcy law, patents, and courts that enforce contracts across state lines. What happened next was not a coincidence.

The killer chart. *The price of American promises* — U.S. government debt traded at a small fraction of face value in the late 1780s; after ratification and Hamilton’s funding plan it approached par (reconstructed from Sylla’s founding-era financial accounts). Paired with a one-slide table of the Constitution’s economic clauses — commerce, contracts, patents, bankruptcy, takings, no state tariffs, no established church — each mapped to the later class where the course meets it again.

The economics. - Commitment device — a constitution binds tomorrow's politicians today. - Common market — no internal tariffs; goods, people, capital move freely. - Public credit — Hamilton's insight that honored debt is cheap debt. - Standardization — one currency and uniform rules cut transaction costs.

The American institution. The most literal class in the course: the Commerce Clause (Class 7's national market), the Patent Clause (Class 8's invention machine), the takings clause (property and land, Class 17), the taxing power (Class 22), and the First Amendment's religious marketplace — the constitutional hooks that every later class hangs on.

Discussion. If you were writing the constitution for a brand-new country today, which economic clauses would you copy from 1787 — and which would you change?

Modern echo. Debt-ceiling standoffs are a live stress test of Hamilton's core achievement: the world's belief that the United States always pays.

Read. The Constitution's economic clauses (Art. I §8, §10; Amendments I and V); Hamilton, *First Report on Public Credit* (excerpt); Federalist No. 11 (excerpt).

Class 6 — Why Nations Diverge: Institutions, Geography, and Economic Beliefs

Part I — The Growth Miracle · *Institution family: growth-enabling*

Institutions appear strongly connected to long-run income differences — but honest economics weighs them against geography, disease, culture, and policy rather than claiming they explain everything.

The hook. Why did some places get rich while others stayed poor? Five explanations compete: geography, disease, natural resources, institutions, and ideas. Which explains the world of 1500? Of 1800? Of 2025? This is a live economics question, not a history lecture.

The killer chart. *The Korean peninsula at night* (NASA, from the International Space Station) — one photo: the South ablaze with light, the North dark; same people, shared history until 1945, radically different rules. Backed by *North vs. South Korea GDP per capita* and the *AJR scatter* plotting protection against expropriation versus income: institutions and prosperity move together.

The economics. - Inclusive vs. extractive institutions — rules that broaden or hoard opportunity. - Natural experiments — borders that isolate rules from geography. - Persistence — institutional shocks echo for centuries. - Zero-sum vs. positive-sum thinking — beliefs shape policy.

The American institution. Broad property ownership, competitive federalism, and open markets were America's inclusive-institution advantage. Honest caveat: slavery and Jim Crow were extractive enclaves inside an inclusive system, and their effects persisted — a tension later classes confront directly.

Discussion. If institutions matter so much, can a country simply choose good institutions? If not, why not?

Modern echo. Chinoy, Nunn, Sequeira & Stantcheva find zero-sum thinking today predicts more support for redistribution and less for immigration — beliefs still shape which rules voters choose.

Read. Acemoglu, Johnson & Robinson (2001), "The Colonial Origins of Comparative Development" (anchor); Diamond, *Guns, Germs, and Steel* (accessible counterpoint).

Class 7 — Land, Rails, and the National Market

Part II — Building the American Growth Machine · *Institution family: market-building*

America became rich partly because it turned a continent into an economy: land institutions that put property into ordinary hands, and networks — railroads, roads, ports, broadband — that fused separated local economies into one enormous integrated market.

The hook. The single most famous photograph in Utah economic history: Promontory Summit, May 10, 1869 — the Golden Spike joins the transcontinental railroad *here*, and a letter from New York to San Francisco goes from months to days. If you grow wheat in Kansas, sell shoes online, or write code in Utah, who is your market? Not just who wants your product — who you can *reach* cheaply enough.

The killer chart. The U.S. railroad map, 1850 vs. 1890, showing a scattered coastal country becoming internally connected in one generation. Paired with the Donaldson–Hornbeck market-access map — the question isn't "does my county have rail?" but "how much does the network improve access to *all* markets?" — read off land values.

The economics. - Market access: how easily a place reaches customers and suppliers. - Network effects: a network gains value as more places connect. - General equilibrium: a railroad changes prices and people everywhere. - Path dependence: connected places compound their advantages for generations.

The American institution. Two institutions built the continental market: land policy — the rectangular survey, the land offices, and the Homestead Act, which moved property from the state into millions of ordinary hands (the transformation from 80 percent of workers on farms to under 2 percent starts here) — and the Commerce Clause's single internal free-trade zone, filled in by a century of network buildout. That is a major reason the U.S. became an industrial and innovation powerhouse rather than a resource exporter.

Discussion. Why might a simple “rail counties are richer” comparison mislead us — and what’s the right counterfactual?

Modern echo. Is lack of broadband today like lack of rail access in 1890 — and what’s the next network: transmission lines, AI data centers, high-speed rail?

Read. Donaldson & Hornbeck (2016, *QJE*), “Railroads and American Economic Growth: A Market Access Approach”; Baum-Snow (2007, *QJE*), “Did Highways Cause Suburbanization?”

Class 8 — The American Invention Machine

Part II — Building the American Growth Machine · *Institution family: market-building*

America’s innovation advantage came from an institutional ecosystem — patents, universities, immigration, science funding, corporate labs, venture capital, large firms, and a culture that rewarded useful tinkering.

The hook. Which invention would you least want to live without? Vote. Now the real question: what kind of society repeatedly *produces, attracts, finances, and scales* such inventions? Why America — not Britain, not Germany, not France? The answer is not one inventor or one law; it’s an ecosystem — and AI is stress-testing it in real time.

The killer chart. U.S. patents, 1790–present, shows the patent system as a central American institution. The Lost Einsteins chart — probability of becoming an inventor by parental income percentile — stuns because talent is everywhere but opportunity to invent is not. The AI investment surge: \$285.9B U.S. private AI investment in 2025, more than 23× China.

The economics. - Nonrivalry: an idea can help every worker at once. - Spillovers: inventors bear costs, society captures the gains. - The patent bargain: temporary monopoly traded for disclosure. - General-purpose technologies: transformation requires complementary investment first.

The American institution. The invention machine touches three institution families: growth-enabling law (the Patent Clause, courts, universities, federal R&D), opportunity-expanding openness (immigration that brought German Jewish chemists — U.S. patenting rose ~31% in émigré fields — and today’s AI researchers), and market-governing tools (DARPA, antitrust). Not laissez-faire, not central planning: a layered ecosystem.

Discussion. If Congress wants to double American innovation, should the next dollar go to patents, universities, immigration, R&D subsidies, prizes, or DARPA-style agencies? Defend with one economic concept.

Modern echo. Is AI more like the light bulb, the internet, the airplane, or the Segway?

Read. Bell, Chetty, Jaravel, Petkova & Van Reenen (2019), “Who Becomes an Inventor in America?”; Moser, Voena & Waldinger (2014), “German Jewish Émigrés and U.S. Invention.”

Class 9 — Factories, Firms, and Scale

Part II — Building the American Growth Machine · *Institution family: market-building*

America’s growth machine did not run on inventions alone: it needed organizations that could coordinate workers, machines, capital, energy, knowledge, logistics, and time at enormous scale.

The hook. Why can McDonald’s make the same fries everywhere? From artisanal workshop to Ford assembly line to AI data center, the common thread is systems for coordinating people, machines, information, and capital. The big idea: organization is itself a technology — coordination, not just invention, is a source of prosperity.

The killer chart. *The price of a Model T* (Ford data via Hounshell) — the list price falls by roughly two-thirds from 1909 to the mid-1920s while annual production explodes from thousands to millions: economies of scale in one line, the chart every student remembers. Supported by the hand-vs-machine labor table from the Attack, Margo, and Rhode analysis of the 1890s Hand and Machine Labor Study — task-by-task evidence that mechanization changed tasks, skills, and scale — and the Census startup-rate chart for the modern business-dynamism discussion.

The economics. - Transaction costs (Coase): using markets isn’t free; sometimes hierarchy is cheaper. - Economies of scale: average cost falls as output rises. - Agency problems: separating ownership from control creates supervision challenges. - Architectural innovation: firms fail when how components fit changes.

The American institution. The corporation — legal personhood, limited liability, contract and bankruptcy law, managerial hierarchy — let American organizations grow enormous without collapsing under their own complexity. The classically liberal takeaway is neither to romanticize small production nor demonize big firms, but to ask what scale does well, what it does badly, and which institutions preserve competition and entry.

Discussion. Is a large firm a threat to capitalism or an achievement of capitalism? (Good answer: both.)

Modern echo. Is an AI lab more like Menlo Park, Bell Labs, Ford, DARPA, or Amazon?

Read. Attack, Margo, and Rhode (2019, *JEP*), “‘Automation’ of Manufacturing in the Late Nineteenth Century”; Chandler (1992, *JEP*), “Organizational Capabilities and the Economic History of the Industrial Enterprise.”

Class 10 — Productivity and Wages: Did the Growth Machine Make Workers Richer?

Part II — Building the American Growth Machine · *Institution family: market-building*

Productivity growth is the central reason workers can become richer over time, but the link from productivity to pay depends on what “pay” means, which workers, what prices they face, and what institutions govern labor markets.

The hook. Would you rather be a typical worker in 1820, 1920, 1970, or today? Ordinary workers are vastly richer than their great-great-grandparents — that’s the big fact. This class rejects two bad stories — “capitalism just exploits workers” and “productivity automatically raises everyone’s wages” — and builds the better economics in between.

The killer chart. Long-run worker compensation, 1790–present (MeasuringWorth), showing the transformation of ordinary workers’ pay across two centuries; paired with BLS labor productivity since 1947, the engine behind it. The famous EPI productivity–pay gap chart appears too — deliberately as a “how to read a chart critically” measurement clinic, not as the thesis.

The economics. - Labor demand: firms pay up to what workers add ($w \approx MRPL$). - Total compensation: wages plus benefits — \$13.79 of \$46.15 hourly. - Mean vs. median: the average and typical worker differ. - Talent misallocation: discrimination shrank the whole pie.

The American institution. America paired an unusually productive growth machine with institutions that translated productivity, imperfectly but really, into worker prosperity: free labor markets, mass public education, immigration, and — over the twentieth century — labor law, unions, taxes, transfers, and social insurance. Pro-growth but not triumphalist: the gains were real, the timing slow, the distribution uneven.

Discussion. The four-charts exercise — which chart would you use if you were a worker, a CEO, a policymaker, a union leader, or an economist?

Modern echo. Hsieh–Hurst–Jones–Klenow show falling discrimination explains 20–40% of U.S. growth since 1960 — opportunity and growth are complements, a live lesson for today’s talent debates.

Read. Stansbury and Summers (2018), “Productivity and Pay: Is the Link Broken?”; Goldin and Margo (1992), “The Great Compression.”

Class 11 — Free and Coerced Labor

Part III — Mobilizing People and Expanding Opportunity · *Institution family: opportunity-expanding*

Part III device: the opportunity ladder — a recurring slide that gains one rung per class (own your labor → bargain → learn → arrive → choose → belong → move → rise) until the full ladder appears in Class 18.

The most basic economic right is ownership of one's own labor; slavery violated that right, generated measured output for enslavers while destroying welfare for the enslaved, and clarifies why free labor is both a moral achievement and a growth institution.

The hook. What do you actually own? Start with a paycheck, a slave schedule listing people as property, and the text of the 13th Amendment. This is not a general history-of-slavery lecture — it is an economics lecture about the foundational question of Part III: who owns the worker?

The killer chart. *Cotton output and slave prices, 1800–1860* — productivity gains were capitalized into the value of enslaved people, proving measured output is not welfare. *Wealth held in enslaved people* — slave capital rose from \$291 million in 1805 to over \$3 billion in 1860, 44 percent of all wealth in the cotton states.

The economics. - Self-ownership — you control your body, labor, and occupation. - Outside option — bargaining power starts with the ability to say no. - GDP vs. welfare — output can rise while human welfare falls. - Talent allocation — exclusion wastes talent; HHJK find it cost real growth.

The American institution. Slavery was property rights in human beings; the 13th Amendment abolished it and made self-ownership the law. Freedpeople's first demands — schools, family, mobility, wage contracts — were precisely the core inputs of opportunity.

Discussion. Can an institution be profitable for some people and harmful to economic development overall?

Modern echo. Hsieh–Hurst–Jones–Klenow find reducing occupational barriers for women and Black Americans explains 20–40 percent of growth since 1960 — today's debates over noncompetes, licensing, and prison labor are debates about exit options.

Read. Gavin Wright, "Slavery and the Rise of the Nineteenth-Century American Economy" (*JEP* 2022); Sokoloff and Engerman, "Institutions, Factor Endowments, and Paths of Development in the New World" (*JEP* 2000).

Class 12 — Labor Markets and Worker Power: Exit, Voice, and Bargaining

Part III — Mobilizing People and Expanding Opportunity · *Institution family: opportunity-expanding*

Free labor markets are among the great institutions of modern prosperity, but wages depend not only on productivity — they also depend on outside options, employer market power, unions, minimum wages, and worker voice.

The hook. Once labor is legally free, how are wages actually set? Reject two bad stories: that workers are always paid exactly what they're worth, and that wages are pure exploitation. Worker power is concrete economics: the ability to say no, leave, bargain, or choose among competing offers.

The killer chart. *Union membership and top-income shares* — the twentieth-century mirror image raises the class's live question: did unions create the mid-century middle class, or grow because it was already rising? *Labor share of output* — a quick reminder of how much of the pie goes to workers as a group.

The economics. - Outside option — what you can do if you leave. - Monopsony — employer power to set wages below workers' contribution. - Exit vs. voice — quit and switch, or bargain and organize. - Causal inference — did the policy cause the outcome, or reflect selection?

The American institution. The NLRA (1935) protects organizing and collective bargaining; the FLSA sets minimum wage and overtime; and federalism lets Utah add its Right to Work Law limiting compulsory union support — a live institutional case students can analyze rather than fight about.

Discussion. Should a worker be required to pay for union representation if the union bargains on that worker's behalf?

Modern echo. Fights over noncompete bans, no-poach enforcement, and \$15+ minimum wages all turn on whether labor markets look competitive or monopsonistic.

Read. Farber, Herbst, Kuziemko, and Naidu, "Unions and Inequality Over the Twentieth Century" (NBER); Card and Krueger's New Jersey–Pennsylvania minimum-wage study.

Class 13 — Education and Human Capital: The American Opportunity Machine

Part III — Mobilizing People and Expanding Opportunity · *Institution family: opportunity-expanding*

Education is one of America's most important opportunity-expanding institutions: it raises human capital, increases workers' outside options, and turns talent into growth — but access, quality, cost, and selection all matter.

The hook. American institutions don't just protect property and build markets — they build people. This is deliberately not a "college is good" lecture: it asks why education pays,

whether the premium is causal, and why America invented mass schooling before anyone else.

The killer chart. *The college wage premium over time* (Goldin–Katz series) — the payoff to a degree fell through mid-century as high schools flooded the market with skills, then roughly doubled after 1980: one line that teaches supply and demand for skill. The BLS *Education Pays* bars (\$738 a week and 6.2 percent unemployment without a diploma, 2024) appear as the setup for the class’s real question: is this causal? *The race between education and technology* completes the story — when skill supply outruns skill demand, inequality can fall.

The economics. - Human capital — productive skills and knowledge inside people. - Signaling — a diploma certifies ability and persistence. - Selection vs. causation — did school cause earnings, or ability? - Credit constraints — people underinvest when they cannot borrow.

The American institution. Goldin argues Americans “invented the publicly funded, academic secondary school for the masses.” The Morrill Act built land-grant universities — Utah State in 1888, alongside the University of Deseret (1850, now the University of Utah) — and the GI Bill put college within reach of millions.

Discussion. Was the high-school movement one of America’s most important economic institutions?

Modern echo. Student debt, low-completion programs, and AI-era reskilling all hinge on whether college still pays — on average, yes; for every program and price, no.

Read. Goldin and Katz, *The Race between Education and Technology*; Oreopoulos and Petronijevic, “Making College Worth It” (NBER).

Class 14 — Immigration and Assimilation: America’s Human-Capital Inflow

Part III — Mobilizing People and Expanding Opportunity · *Institution family: opportunity-expanding*

Immigration has been one of America’s major growth institutions — expanding labor, entrepreneurship, innovation, and pluralism — while also generating real distributional pressures, assimilation challenges, and political backlash.

The hook. Education builds human capital at home; immigration imports it from the world. The class teaches students the economist’s question about any immigration claim: compared with what counterfactual, over what time horizon, and for whom?

The killer chart. *Foreign-born share of the U.S. population, 1850–present* — 50.2 million immigrants in 2024, 14.8 percent of the population, exactly matching the 1890 peak: the

number is unprecedented, the share is not. *CBO's immigration GDP effect* — the 2021–2026 surge projected to add \$8.9 trillion to GDP over 2024–2034 and cut deficits by \$0.9 trillion.

The economics. - Complements vs. substitutes — immigrants compete with some workers, boost others. - Capital adjustment — firms expand when labor supply grows. - Assimilation — measurable integration in earnings, language, and citizenship. - Counterfactual — what would have happened without the inflow?

The American institution. Immigration policy defines who may join the growth machine — from the Chinese Exclusion Act (1882) and 1924 quotas to naturalization itself. Religious liberty is part of the story: by 1900, roughly 50,000 LDS converts from the British Isles and 30,000 from Scandinavia had immigrated to the Intermountain West, with churches reducing migration costs and supporting assimilation.

Discussion. Why can immigration create political backlash even when it does not lower native wages?

Modern echo. The post-2021 immigration surge poses exactly the class's question: national GDP gains versus concentrated local costs in housing, schools, and budgets.

Read. Abramitzky and Boustan, *Streets of Gold*; Card, "The Impact of the Mariel Boatlift on the Miami Labor Market."

Class 15 — Women, Work, and Family: The Household Economy Enters the Market

Part III — Mobilizing People and Expanding Opportunity · *Institution family: opportunity-expanding*

The twentieth-century rise of women's education and market work dramatically expanded America's effective human capital, while changing marriage, fertility, household production, and the meaning of work–family choice.

The hook. A parent caring for a child adds nothing to GDP; paid daycare does. Women's economic history is partly a measurement problem — and women's rising education, work, and control over fertility produced one of the largest reallocations of talent in American history. Yet the household did not disappear.

The killer chart. *Women's labor-force participation, 1890–present* — Goldin's U-shape shows growth did not automatically raise women's market work. *The gender wage gap, 1960–present* — the ratio rose from 75 percent in 2000 to 82 percent in 2024; the raw gap is a fact, not an explanation. *U.S. total fertility rate since 1909* — decline, baby boom, modern lows.

The economics. - Household production — valuable work GDP never counts. - Opportunity cost — the value of the next best use of time. - Child penalty — mothers' earnings dip after children arrive. - Quantity–quality tradeoff — fewer children, more investment per child.

The American institution. The Married Women's Property Acts gave married women rights to own property and keep earnings — a concrete institutional transformation, extended by suffrage, anti-discrimination law, and equal credit access. Religious and family institutions shape norms and social capital and deserve respectful economic analysis, which plays honestly in Utah — where the local hook writes itself: Utah, long America's highest-fertility state, has now fallen below replacement.

Discussion. Should low fertility be treated as a problem — and why is it falling even in Utah?

Modern echo. Falling fertility drives 2020s debates over Social Security, childcare costs, immigration, and pro-family policy.

Read. Goldin, “The Quiet Revolution That Transformed Women's Employment, Education, and Family” (*AER* 2006); Goldin, “A Grand Gender Convergence: Its Last Chapter” (*AER* 2014).

Class 16 — Race, Migration, and Opportunity: When Institutions Exclude Talent

Part III — Mobilizing People and Expanding Opportunity · *Institution family: opportunity-expanding*

Race shaped American economic outcomes because laws, violence, segregation, and exclusion restricted access to the growth machine; when those barriers weakened, more talent flowed to productive uses, raising both opportunity and output.

The hook. This is an economics lecture, not a guilt lecture. Exclusion is not only unjust — it is economically destructive, because it blocks people from education, occupations, property, credit, and trust. A pro-growth society should care deeply about removing artificial barriers to talent.

The killer chart. *The Black–white male earnings gap, 1940–2014* — Bayer and Charles show it narrowed to the mid-1970s and then widened again; the puzzle organizes the whole class. *Black population by region, 1900 vs. 1970* — the Great Migration as one of the largest internal reallocations of labor in U.S. history.

The economics. - Migration as exit — moving to escape bad local institutions. - Talent misallocation — blocked people, lost output for everyone. - Political representation —

Black officeholders raised local public spending. - Trust — Tuskegee’s disclosure cost Black men measurable life expectancy.

The American institution. The 13th, 14th, and 15th Amendments, the Civil Rights Act, and the Voting Rights Act are economic institutions defining access to labor, schools, housing, and courts. The American achievement is not that exclusion never existed — it is that the rules could be contested, amended, and reformed toward broader participation.

Discussion. Is migration enough to create opportunity?

Modern echo. Alsan–Wanamaker’s finding that institutional betrayal breeds costly, rational mistrust speaks directly to post-pandemic debates over public health and confidence in institutions.

Read. Bayer and Charles, “Divergent Paths”; Derenoncourt, “Can You Move to Opportunity? Evidence from the Great Migration” (*AER*).

Class 17 — Cities, Housing, and Path Dependence: Place as an Institution

Part III — Mobilizing People and Expanding Opportunity · *Institution family: opportunity-expanding*

Cities create opportunity by concentrating workers, firms, ideas, and markets, but housing supply, zoning, segregation, and path dependence determine who can actually access that opportunity.

The hook. Two nearly identical houses, wildly different prices — what exactly are people buying when they buy location? Place is not just geography; it is a bundle of labor markets, schools, housing rules, networks, and history. Opportunity has an address.

The killer chart. *Home price-to-income ratios* — the median home hit five times median household income in 2024, versus 3.2 in the 1990s; if productive cities pay better wages, why don’t workers move? Housing eats the gains. *Urbanization since 1790* — growth is also spatial transformation. *Real home prices since 1890 (Shiller)* — the modern break from a long history of affordability.

The economics. - Agglomeration — people and firms are more productive near each other. - Capitalization — a location’s benefits show up in its prices. - Path dependence — early advantages stick; history shapes today’s map. - Spatial misallocation — blocked migration makes the whole country poorer.

The American institution. Zoning is itself a set of property rules — local, legal, and political — and federalism means local rules shape national opportunity. Utah makes it vivid: the Gardner Policy Institute ranked Utah the ninth-most expensive housing market in 2024, with a median single-family price of \$547,700.

Discussion. Should high-opportunity cities have an obligation to build more housing?

Modern echo. The 2020s YIMBY-versus-local-control fight — including Utah’s own debates over starter homes and density near transit — is Ganong–Shoag and Hsieh–Moretti made political.

Read. Bleakley and Lin, “Portage and Path Dependence” (*QJE* 2012); Hsieh and Moretti, “Housing Constraints and Spatial Misallocation” (*AEJ: Macro*).

Class 18 — The American Dream and Mobility: Does Growth Let People Rise?

Part III — Mobilizing People and Expanding Opportunity · *Institution family: opportunity-expanding*

The American Dream depends on both growth and opportunity: growth raises the ceiling for everyone, but mobility depends on how education, neighborhoods, family, networks, and institutions shape children’s chances.

The hook. Open with a poll: will you out-earn your parents? For Americans born in 1940 the answer was yes about 90 percent of the time; for those born in the 1980s, about 50 percent. Then the opportunity ladder appears complete — all eight rungs built across Classes 11–17 — and the capstone question lands: growth is necessary but not sufficient for the American Dream.

The killer chart. *Share of children earning more than their parents, by birth cohort* — Chetty et al.’s fading-dream chart, falling from ~90 to ~50 percent; distributing growth more equally would reverse more than 70 percent of the decline.

The economics. - Absolute mobility — children earning more than their parents. - Relative mobility — moving up the ladder compared with others. - Exposure effects — more childhood years in a good place matter. - Economic connectedness — cross-class friendships predict rising incomes.

The American institution. The Dream is not one national fact but a local institutional outcome: bottom-to-top mobility is 4.4 percent in Charlotte but 12.9 percent in San Jose. The Opportunity Atlas — shown for Salt Lake County for local punch — reveals outcomes varying sharply across nearby neighborhoods, and Utah’s families, churches, and voluntary associations illustrate why economists now study social capital.

Discussion. Should policy move poor families to opportunity, or bring opportunity to poor places?

Modern echo. Housing vouchers, exclusionary-zoning reform, and place-based policy are the live 2020s versions of the move-people-versus-bring-opportunity tradeoff.

Read. Chetty, Grusky, Hell, Hendren, Manduca, and Narang, “The Fading American Dream” (*Science*); Chetty et al., “Social Capital I” (*Nature*).

Class 19 — Trade, Tariffs, and Globalization: When the Market Becomes Global

Part IV — Governing Capitalism’s Consequences · *Institution family: market-governing*

Trade raises living standards through specialization, competition, and lower prices, but it creates concentrated adjustment costs and security risks that American institutions manage through tariffs, agreements, worker-adjustment policy, and strategic restrictions.

The hook. Where was your morning made? Your phone, your clothes, your breakfast crossed oceans to reach you. Trade makes America richer overall — yet it can hollow out particular towns, and that tension is where Part IV begins.

The killer chart. *U.S. trade as a share of GDP, 1790–present* — globalization isn’t new, but its scale and speed are. *Manufacturing employment by import exposure across commuting zones* — the China shock made towns concentrated in import-competing manufacturing measurably, persistently worse off.

The economics. - Comparative advantage — specialize in what you give up least. - Tariff incidence — who actually pays the import tax. - Adjustment costs — workers and towns can’t instantly move or retrain. - Resilience — insurance against supply shocks isn’t free.

The American institution. Trade policy is one of the oldest economic-policy debates in U.S. history, running through the customs system, the GATT/WTO order, Section 232 and 301 tariffs, Trade Adjustment Assistance, and CHIPS-style industrial policy. Free trade is not the absence of institutions — it requires ports, courts, customs enforcement, and rules for national security.

Discussion. A ten-minute comparative-advantage trading game — students get random endowments and trade — then the debate question: are tariffs patriotic?

Modern echo. The 2020s tariff wars and semiconductor restrictions ask whether national security justifies departures from free trade — and when it’s just protectionism in uniform.

Read. Autor, Dorn, and Hanson, “The China Syndrome” (*AER*); Douglas Irwin, *Peddling Protectionism: Smoot–Hawley and the Great Depression*.

Class 20 — Market Power and Antitrust: When Big Firms Help, and When They Harm

Part IV — Governing Capitalism’s Consequences · *Institution family: market-governing*

Antitrust is an American institution designed to preserve capitalism by protecting competition: large firms can create enormous value, but market power can raise prices, suppress wages, block entry, and distort politics.

The hook. Would you break up Amazon? Before you answer, notice the paradox: the same scale that delivers two-day shipping and cheap cloud computing can also block rivals and squeeze suppliers. The antitrust question isn’t “big or small?” — it’s “competitive or protected?”

The killer chart. *The rise of markups, 1955–present* — De Loecker, Eeckhout, and Unger find prices above cost jumped from about 21 percent to 61 percent after 1980, driven by the top firms. *The declining U.S. startup rate* — a dynamic economy needs entry, and entry is falling.

The economics. - Market power — ability to influence prices, wages, or entry. - Monopsony — buyer power that pushes wages down. - Network effects — products worth more as users multiply. - Contestability — can new entrants challenge incumbents?

The American institution. The Sherman Act (1890), Clayton Act, and FTC Act built a distinctly American answer: not socialism, but rules that keep markets open and contestable. From Standard Oil (1911) to AT&T to Microsoft, American capitalism has never meant “firms can do anything” — it has meant markets under rules that protect competition, not competitors.

Discussion. Is Amazon more like Standard Oil, Walmart, a mall, a railroad, or a government?

Modern echo. The Google, Apple, Amazon, and Meta cases are a live laboratory for whether century-old antitrust tools fit platform economies.

Read. De Loecker, Eeckhout, and Unger, “The Rise of Market Power” (*QJE*); FTC, “The Antitrust Laws.”

Class 21 — The Useful State: Public Goods, Infrastructure, and State Capacity

Part IV — Governing Capitalism’s Consequences · *Institution family: market-governing*

A successful market economy needs a capable but limited state: government is most justified when it solves specific collective-action problems markets

underprovide — defense, courts, infrastructure, public health, basic research — judged always by costs, incentives, and evidence.

The hook. Would you voluntarily pay for national defense if your neighbors got it free either way? Nobody would — and that free-rider problem explains why some valuable things never get built by markets alone. But “this is important” is not an economic case for government; “markets have a specific failure here” is.

The killer chart. *Federal spending as a share of GDP, by category* — the federal government spent \$7.01 trillion in FY2025, about 23 percent of GDP, and most of it isn’t roads or schools but social insurance, health, defense, and interest. That gap between theory and budget stuns students.

The economics. - Public good — nonrival and nonexcludable, like defense or knowledge. - Free-rider problem — people benefit without paying, so too little gets produced. - State capacity — government’s ability to tax, build, and enforce well. - Government failure — public action distorted by capture, incentives, and politics.

The American institution. This class asks what the American state is *for*: courts and rule of law, public-health agencies, land-grant universities, NSF/NIH/DARPA, and a federalist structure that pushes most provision to states and localities. Religious liberty matters here too — churches and mutual-aid societies provide local public goods outside the state, a very Utah-relevant point.

Discussion. If something is socially valuable, should government provide it?

Modern echo. Rural broadband, pandemic preparedness, and infrastructure bills all turn on the same question: real market failure, or just political spending?

Read. Coase, “The Problem of Social Cost”; CDC, “Ten Great Public Health Achievements.”

Class 22 — Fiscal Capacity: Taxes, Insurance, Debt, and the Cost of Promises

Part IV — Governing Capitalism’s Consequences · *Institution family: market-governing*

Fiscal capacity is a core American institution: the U.S. became powerful partly because it could tax and borrow credibly, but today social-insurance promises, health spending, interest costs, and aging are outgrowing the revenue system built to support them.

The hook. Who pays for the promises? Every Social Security check, Medicare payment, and interest dollar has to come from taxes today, taxes tomorrow, inflation, cuts, or growth. There is no sixth option — and you, the 18-year-old in this room, are the taxpayer of tomorrow.

The killer chart. *Federal debt held by the public, 1790–2055* — CBO projects debt reaching its highest level ever in 2029 and 156 percent of GDP by 2055. *Net interest* — rising from 3.2 to 5.4 percent of GDP, projected to exceed all discretionary spending by 2052: past promises crowding out future choices.

The economics. - Tax incidence — who actually bears the burden after prices adjust. - Social insurance — pooling risks over life and across people. - Deficit vs. debt — annual flow versus accumulated stock. - r vs. g — interest rates racing economic growth.

The American institution. Hamilton's 1790 assumption of state debts built American public credit deliberately, and it made the U.S. able to fight wars, stabilize crises, and sustain the world's most important sovereign-debt market. Treasury, the IRS, Social Security, Medicare, and CBO carry that institution today.

Discussion. Should fiscal policy prioritize low taxes, generous benefits, or low debt — and why can you only pick two?

Modern echo. The Social Security and Medicare trust funds are both projected depleted in 2033 — well within students' working lives.

Read. CBO, *The Long-Term Budget Outlook: 2025 to 2055*; Hamilton, *First Report on Public Credit* and the Compromise of 1790.

Class 23 — Money, Banking, and Crises: Fragility, the Fed, and the Lender of Last Resort

Part IV — Governing Capitalism's Consequences · *Institution family: market-governing*

Modern capitalism needs banks because they turn savings into credit and liquidity, but banks are inherently fragile because they borrow short and lend long; central banks and deposit insurance exist to stop private fear from becoming economy-wide collapse.

The hook. What makes a dollar worth anything? Only the belief that everyone else will accept it — money is trust made transferable. And your bank isn't a vault: your deposit is already lent out as someone's mortgage, which is exactly why banks are useful and exactly why they can fail.

The killer chart. *The U.S. banking-panics timeline* — 1837, 1857, 1873, 1893, 1907, 1930–33, 2008, 2020, 2023: panics are not one-off accidents but a recurring feature of American finance. *Inflation, unemployment, and the federal funds rate* — the whole modern monetary-policy story in one picture.

The economics. - Maturity transformation — borrowing short to lend long. - Bank run — fear can create the thing feared. - Lender of last resort — lend freely, on good collateral, at a penalty rate. - Moral hazard — rescues today can invite recklessness tomorrow.

The American institution. America abolished one central bank, resisted another for decades, then built the Federal Reserve after the Panic of 1907 exposed the absurdity of relying on J.P. Morgan to stop a national panic. The Fed's failure in the early 1930s — the money supply fell nearly 30 percent — produced the FDIC in 1933, and 2008 showed run logic migrating into shadow banking.

Discussion. The classroom bank-run game: students are depositors who choose each round whether to withdraw; watch a healthy bank die from fear alone — the most memorable fifteen minutes in the course. Then: why are banks fragile even when bankers are honest?

Modern echo. The 2023 bank failures and the 2021–23 inflation fight showed both faces of the Fed — fire department and demand manager — within two years.

Read. Diamond and Dybvig, “Bank Runs, Deposit Insurance, and Liquidity”; Bernanke, “Money, Gold, and the Great Depression.”

Class 24 — Health, Environment, and Externalities: Harms and Benefits Outside GDP

Part IV — Governing Capitalism's Consequences · *Institution family: market-governing*

Many of capitalism's biggest costs and benefits occur outside market prices — pollution, disease, safety, longevity; economics gives us tools to measure those externalities and design institutions that reduce harms without pretending regulation is free.

The hook. GDP counts the ambulance ride, but not the accident avoided. A factory's output raises GDP; the asthma its smoke causes doesn't lower it — and the inhalers people buy raise GDP again. If that sounds broken, this class is about how economists fix the accounting.

The killer chart. *Air pollution falls while GDP rises* — EPA reports the six common pollutants fell 79 percent from 1970 to 2024 while the economy grew: America didn't choose between growth and clean air. *U.S. spends more, lives shorter* — life expectancy hit 79.0 years in 2024, still 3.7 years below comparable countries despite the world's highest health spending.

The economics. - Externality — private choices imposing unpriced costs or benefits. - Pigouvian tax — pricing the harm to shrink it. - Value of a statistical life — society's willingness to pay for risk reduction. - Health production — medical care is only one input into health.

The American institution. The Clean Air Act is a signature market-governing achievement — standards, monitoring, and enforcement whose benefits EPA estimates exceeded costs

by more than 30 to 1. The EPA, CDC, FDA, OSHA, and tort law make markets more complete by pricing or preventing harms markets ignore.

Discussion. Should economists put a dollar value on life?

Modern echo. Climate policy is the hardest externality yet — global, long-lived, and intergenerational, where local-pollution tools don't simply transfer.

Read. Arrow, “Uncertainty and the Welfare Economics of Medical Care”; EPA, *Our Nation's Air 2025*.

Class 25 — AI and the Institutions of Progress: Can America Keep Producing Broad-Based Prosperity?

Part IV — Governing Capitalism's Consequences · *Institution family: market-governing*

The final question is not whether AI raises GDP, but whether American institutions can convert rapid idea-driven progress into broad-based welfare — while managing physical-world bottlenecks, inequality, and catastrophic risk.

The hook. The course ends by turning its own question forward. America built a growth machine out of institutions — rights, markets, schools, firms, finance, competition, correction. AI is the biggest stress test that machine has faced in a century. Can it do it again?

The killer chart. *Fast America, slow America* (BLS data + Goolsbee–Syverson) — U.S. construction productivity has *fallen* for fifty years while the software economy soared: the same country that trains frontier AI models cannot build housing, transmission lines, or nuclear plants any faster than in 1970. Supported by *global corporate AI investment* — U.S. private AI investment hit \$285.9 billion in 2025, more than 23 times China's reported total, with 1,953 newly funded American AI companies.

The economics. - General-purpose technology — an invention that reorganizes the whole economy. - Complementary investment — AI matters only when workflows actually change. - Baumol bottleneck — the slowest sectors come to dominate costs. - Speed vs. risk — move fast where delay is costly, carefully where mistakes are catastrophic.

The American institution. Technology works *through* institutions: universities, immigration, venture capital, antitrust, the FDA, permitting agencies, schools, and civil society will decide who benefits from AI and biotech. America's paradox is that it can invent in the world of bits — mRNA vaccines in months under Operation Warp Speed — while construction productivity has fallen for decades, leaving housing, transmission, and nuclear power stuck in slow America.

Discussion. Is AI more like electricity, the internet, or the Segway?

Modern echo — and the recruiting close. Every question in this class is an open research question that economists are working on right now. The last slide is the pitch made explicit: if this course was interesting, this is what the economics major *is* — take ECON 2010 next, look at the QAMO major, and here is what economics graduates do and where they go.

Read. Stanford HAI, *AI Index Report 2026*; Goolsbee and Syverson, “The Strange and Awful Path of Productivity in the U.S. Construction Sector.”

The closing argument. Growth creates possibility; institutions determine whether possibility becomes prosperity; welfare asks whether prosperity becomes longer, healthier, freer, more meaningful lives. For most of history, ordinary people were poor, sick, and short-lived — then growth happened, and it was never just machines or money. It was ideas, markets, law, schools, firms, immigration, finance, competition, and the repeated correction of capitalism’s failures. Whether AI and biotech produce broad-based progress depends on whether American institutions can keep adapting — and that is the question this course hands to its students.